Binary Sorting

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

You are given a binary string of length N that consists only of `0` and `1`. You may perform the following operation any number of times.

- Choose two indices i and j (where $i \neq j$ and $1 \leq i, j \leq N$), and swap the characters at positions i and j.

The goal of the task is to sort the string (i.e., every `0` appears before every `1`). Find the minimum number of swap operations required to sort the string. You must also output a sequence of that many swaps that sorts the string.

Input

The first line contains an integer N — the length of the binary string.

The second line contains a string of length N consisting only of `0` and `1`.

Output

On the first line, output an integer K — the minimum number of swap operations required to sort the string.

Then, output K lines. The t-th of these lines contains two space-separated integers i and j — the indices swapped in the t-th operation. The operations are applied in this order.

If there are multiple answers, output any of them.

Scoring

For all tests, the input will satisfy the following constraints:

- $1 \leq N \leq 2 \times 10^5$
- The string consists only of `0` and `1`.

Output Task

Subtask	Output file(s)	Points	Additional constraints
0	`out_sample_01.txt` `out_sample_02.txt`	0	Samples.
1	`out_01.txt`	5	The string contains exactly one `1`.
2	`out_02.txt`	5	$N = 10$
3	`out_03.txt`	5	
4	`out_04.txt`	5	Every `1` appears before every `0`.

Code Task

Subtask	Points	Additional constraints
0	0	Samples.
1	20	Every `1` appears before every `0`.
2	30	$N \leq 10^4$
3	30	—

Examples

standard input	standard output
6 010100	2 2 6 4 5
5 00111	0

Note

Sample 1

This sample is valid for code subtasks 2 and 3.

The string is `010100`. After the first operation ($i = 2$, $j = 6$), the string becomes `000101`. After the second operation ($i = 4$, $j = 5$), the string becomes `000011` and is sorted.

It can be shown that it is impossible to sort the string using only one swap operation.

Swapping positions 5 and 2, then positions 4 and 6, is also a valid answer.

Sample 2

This sample is valid for code subtasks 2 and 3.

The string is already sorted, so $K = 0$ and no swaps are output.